A 13-ENOB Second-Order Noise-Shaping SAR ADC Realizing Optimized NTF Zeros Using the Error-Feedback Structure

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***Abstract*— The noise-shaping successive approximation register (NS-SAR) analog-to-digital converter (ADC) is an emerging hybrid architecture that achieves high resolution and power efficiency simultaneously by combining the merits of the SAR ADC and the $\Delta\Sigma$ ADC. Most prior works adopting the cascaded integrator feed-forward (CIFF) structure demonstrate inefficiency in realizing optimized noise transfer function (NTF). This paper presents a second-order NS-SAR ADC employing the error-feedback (EF) structure to realize complex NTF zeros for noise-shaping enhancement with the minimum modification to a standard SAR. It implements a low-power scaling-friendly EF path by using a passive finite impulse response (FIR) and a comparator-reused dynamic amplifier with process-voltage-temperature (PVT) tracking background calibration. Fabricated in 40-nm CMOS, the prototype chip consumes 84 μW when operating at 10 MS/s. The NS-SAR achieves peak Schreier FoM of 178 dB with 79-dB signal to noise and distortion ratio (SNDR) at an oversampling ratio (OSR) of 8.**

***Index Terms*— $\Delta\Sigma$ modulator, analog-to-digital converter (ADC), digital background calibration, dynamic amplifier, error-feedback (EF) structure, noise coupling, noise shaping, oversampling ADC, successive approximation register (SAR).**

I. Introduction

THE boom of Internet of Things (IoT) is driving the world into a ubiquitous sensing environment with an emphasis on edge computing. This regime brings stringent requirements for both high resolution and low power to its analog-to-digital converters (ADCs), as to satisfy the need for digitizing diverse high-dynamic-range (DR) sensor outputs under a tight power budget. The successive approximation register (SAR) ADC is regarded as a candidate architecture, for its superior power efficiency and simple mostly digital implementation. However, it is challenging to extend these advantages to high-resolution design owing to limitations from comparator noise and excessive digital-to-analog converter (DAC) power/area. $\Delta\Sigma$ ADCs are preferred choices from the high-resolution perspective in light of their noise-shaping capability. Nevertheless, their reliance on operational transconductance amplifiers (OTAs) places bottlenecks on power efficiency and scaling friendliness.

To address these challenges, diverse techniques have been explored to improve the resolution of SAR ADCs or reduce the power of $\Delta\Sigma$ ADCs. On the SAR side, majority voting and statistical estimation are proposed in [1] and [2], respectively, to suppress comparator noise using the hidden information in the residue voltage. Nevertheless, Harpe *et al.* [1] necessitate a carefully tuned meta-stability detector, and Chen *et al.* [2] suffer from severe speed penalty due to requiring many repeating least significant bit (LSB) comparisons. Pipelining is another solution to relaxed the noise constrain by leveraging interstage gain [3]–[7]. However, they are sensitive to the interstage gain error that causes noise leakage from the first stage. On the $\Delta\Sigma$ side, voltage controlled oscillator (VCO)-based integrators and quantizers are proposed to replace their power hungry OTA-based counterpart [8]–[11]. However, extra circuit complexity is also introduced to address the process-voltage-temperature (PVT) sensitivity and non-linearity of the VCO.

Recent research has brought forward the noise-shaping SAR (NS-SAR) ADC as an attractive solution. This emerging architecture hybridizes the energy-efficient SAR operation with the $\Delta\Sigma$ mechanism, fusing their respective merits to overcome the other's limitations in noise/power, thus unveiling great potential for co-realization of high resolution and low power. The NS-SAR extends the operation of a standard SAR ADC with a few extra steps that filter and feed back previous conversion residues to enable the noise-shaping effect, as illustrated in Fig. 1(a). These steps, in analogy to the loop filters in $\Delta\Sigma$ ADCs, are the major factors governing the overall performance of the NS-SAR ADC. References [12]–[14] adopted active-*RC* integrators in their loop filters to obtain relatively sharp noise transfer function (NTF). However, the involvement of OTAs deteriorated power efficiency and scaling friendliness of the SAR ADC. Chen *et al.* [15], [17], Guo and Sun [16], Guo [18], and Lin it et al. [19] demonstrated noise shaping in various orders using fully passive switched-capacitor (SC) filters. These designs maintained a fully dynamic manner, thus can be easily duty-cycled and scaled. However, their lossy filters rendered relatively shallow NTF and rise of thermal noise, leading to DR degradation. Liu and Huang [20] and

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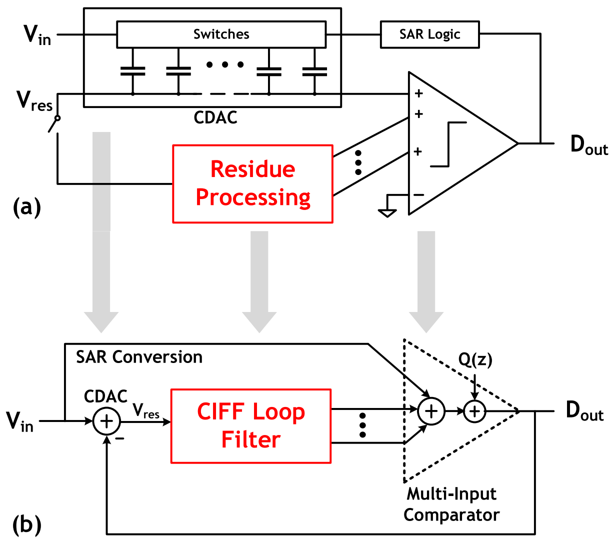

Fig. 1. (a) Abstracted block diagram of existing NS-SAR implementations and (b) their mapping to the CIFF structure.

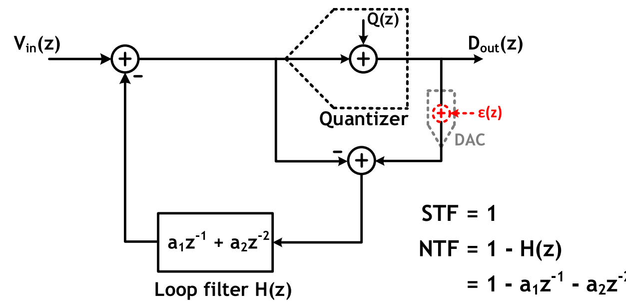

Fig. 2. Generic signal flow diagram of the EF structure.

Miyahara and Matsuzawa [21] combined the passive SC filters with dynamic amplifiers (D-Amps) in their loop filters to co-optimize noise suppression and power. Nevertheless, they became less robust due to the D-Amp's vulnerability to PVT variations.

Despite implementing the loop filter in different circuit topologies, a deeper study reveals that most of the existing designs adopted the cascaded integrator feed-forward (CIFF) structure, as shown in Fig. 1(b). The SAR conversion corresponds to the input direct feed-forward path and the feedback summing node in this structure. Apart from the topology-specific tradeoffs, they also share in common the following limitations on the architectural level. First and foremost, no prior NS-SAR has yet realized NTF zero optimization, which is a common practice in $\Delta\Sigma$ ADCs that maximally attenuate in-band quantization noise. It is a desirable feature to include for better performance. However, under CIFF structures, achieving effective NTF optimization demands the loop filter poles to be in close proximity to the unit circle. It hence necessitates low-loss integrators such as active-*RC* and implying high power and design complexity. Second, as illustrated in Fig. 1, prior design adopting the CIFF structure relied on using multi-input comparators to perform the feed-forward path summing. Despite being convenient, the extra input pairs contribute additional noise and mismatch, leading to potential performance loss.

This paper seeks to further advance the performance of the NS-SAR ADC over prior designs, with the goal to attain strong noise-shaping effect, low power consumption, and PVT robustness simultaneously. The above-mentioned review inspires us to explore from an architectural perspective. Instead of adopting the CIFF structure, this paper presents a second-order NS-SAR ADC through exploiting the error-feedback (EF) structure. Its key technical contributions over prior arts lie in demonstrating: 1) an architecture capable of realizing optimized complex NTF zeros efficiently by simply using charge sharing summation, a passive SC finite impulse response (FIR) filter, and a comparator-reused D-Amp and 2) a reduced-variable EF scheme to simplify the FIR matching requirement and facilitate the flexible tuning of the NTF notch via the D-Amp.

The proposed design requires a minimum modification to a standard SAR ADC, and thus inherits the merits of dynamic operation and hardware simplicity, allowing fast design turn-around.

To mitigate the PVT sensitivity of the D-Amp, this paper develops a dither-based mixed-signal background gain calibration scheme. Background calibration is traditionally deemed non-trivial for $\Delta\Sigma$ ADCs, as it relies on accurately replicating the analog transfer function in the digital domain. Our scheme takes advantage of the simplicity of the EF structure, i.e., its analog transfer function is well-controlled and easy to be digitally represented, to enable precise gain tracking using low-power digital techniques. It provides a practical calibration solution in the context of NS-SAR, and demonstrates a good compatibility with today's growing calibration practice in the industry.

This paper is an extension of the work reported in [22] and is organized as follows. In Section II, we will briefly review the EF structure and prior attempt for building EF NS-SAR. Section III will analyze the architectural features and considerations of the proposed EF NS-SAR. Section IV presents the circuit implementation of the ADC. The measurement results of the prototype and comparison with state of the arts follow in Section V. We then brought up the conclusion in Section VI.

II. Recapitulate the EF Structure

To provide a foundation for the proposed work, this section briefly reviews the essentials of the EF structure. Fig. 2 depicts a generic block diagram of the EF structure. In contrast to the CIFF methods that process the shaped quantization noise, the EF path extracts and filters the unshaped quantization error $Q(z)$. The filtered prior errors are then combined with the input signal during the subsequent quantization. This structure produces a set of the neat signal transfer function (STF) and NTF as shown in Fig. 2. The NTF expression indicates an attractive feature, that using an FIR as the loop filter, rather than low-loss integrators, will suffice for creating highly effective optimization notches. To elucidate with a second-order example, a two-tap FIR filter $(a_1z^{-1} + a_2z^{-2})$ results in an NTF of $1 - a_1z^{-1} - a_2z^{-2}$, whose zeros are at $z = ((a_1 \pm (a_1^2 + 4a_2)^{1/2})/2)$. Provided that $a_2 = -1$ and

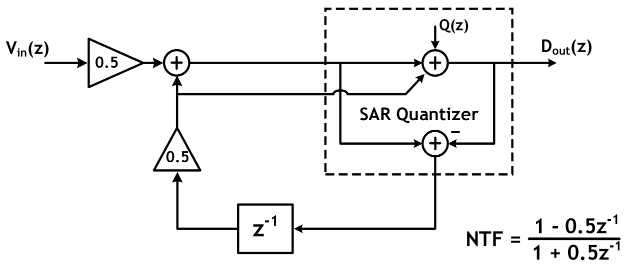

Fig. 3. Architectural block diagram of [15].

$a_1 \leq 2$, the NTF zeros will be placed on the unit circle, thus realizing a strong notch. FIR filters are simply combinations of delay elements. They are relatively easy to design, thus allowing the EF structure to achieve aggressive noise shaping more efficiently. This merit makes the EF structure dominant in digital $\Delta\Sigma$ modulators.

In classic $\Delta\Sigma$ ADCs, however, this advantage of the EF structure is overshadowed by its sensitivity to error [$\varepsilon(z)$ in Fig. 2] in the quantization residue extraction. This error originates from the reference level mismatch between the quantizer (often a flash ADC) and the extraction DAC and usually appears at the output unattenuated. Owing to this drawback, the EF structure used to only function as an order booster at the backend of a high-order integrator-based loop where its non-idealities causes little impact [23], [24], thereby much restricting its potential. In the context of NS-SAR, on the other hand, this limitation is naturally resolved. The SAR operation concurrently quantizes and extracts using the same DAC, thus inherently providing accurate residues at the end of each conversion, allowing the EF structure to be adopted straightforwardly and reliably. This key observation motivates our efforts to advance the NS-SAR by exploiting the architectural advantage of the EF structure.

The first attempt to use the EF structure for NS-SAR was reported by Chen *et al.* [15], who proposed a modified EF scheme to suit their fully passive implementation, as illustrated in Fig. 3. Nevertheless, their scheme only provides the first-order shaping with a maximum of 9.5-dB in-band attenuation. It also suffers from significant signal attenuation due to internal charge sharing, which results in limited DR performance. In connection with [15], our design provides a few new perspectives on EF NS-SAR.

1) It offers better efficiency for high resolution needs by providing optimized higher order NTF.
2) It actualizes a background calibration scheme and allows more reconfiguration flexibility.

III. Proposed EF-NSSAR Architecture

A. System Overview

Fig. 4 depicts the architectural block diagram of the proposed second-order EF NS-SAR ADC. Our work adopts the "D-Amp plus passive SC" approach [20], [21] in implementing the EF path. In analogy to a pre-amplifier, by providing gain at the front of the EF operation, the D-Amp suppresses the noise contributed by the rest EF path. This allows relatively small capacitors to be used for the EF filter, which curtails the area overhead. More importantly, with the path gain supported by the D-Amp, we implement the FIR EF filter as fully passive, and the subsequent feedback summation in the form of charge sharing [i.e., merging the FIR capacitors with the SAR's capacitor DAC (CDAC)] without penalty in the noise-shaping effect. In this way, the effective noise shaping is realized in a highly simple and low-power manner, hence fully exerting the potency of the EF structure. To mitigate the PVT sensitivity of the D-Amp, this paper develops a low-cost background calibration scheme, whose discussion follows in Section IV.

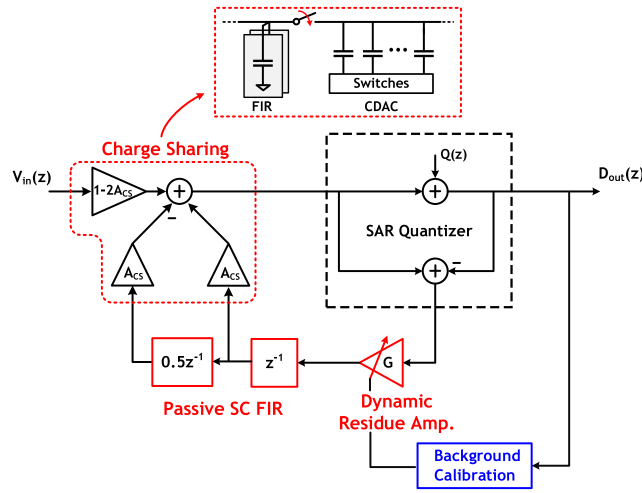

Fig. 4. Architectural block diagram of the proposed second-order EF NS-SAR.

Noteworthily, the charge sharing summation transfers all the necessary information to the capacitor array for the SAR operation to quantize and extract the next residue accurately. Therefore, unlike prior works requiring a multi-input comparator, this paper only needs a standard two-input comparator, thus implicitly reducing the number of noise and mismatch sources. CIFF-based designs, on the other hand, do not share this benefit, because merging the loop filter output with the CDAC results in information loss as the input signal and the filtered noise become inseparable, making the residue extraction incorrect.

B. Reduced-Variable EF Scheme

To further facilitate a robust design, this paper proposes a reduced-variable EF scheme. As suggested in Fig. 4, instead of relying on both the magnitude and the ratio of the two FIR coefficients to define the NTF zeros (as explained in Section II), this design keeps a fixed FIR coefficient ratio in integer, and use the EF path gain K_{EF} as the sole variable to modulate the NTF. Our NTF is therefore expressed as

$$\mathrm{NTF}(z) = 1 - K_{\mathrm{EF}}(z^{-1} - 0.5z^{-2}) \tag{1}$$

where the EF path gains $K_{\mathrm{EF}} = GA_{\mathrm{CS}}$, G is the D-Amp gain, and A_{CS} is the charge sharing attenuation factor. The benefit of this scheme is twofold. First, it simplifies the design of the FIR and guarantees better matching, since the integer ratio can be easily implemented with identical unit cells. Second, and more

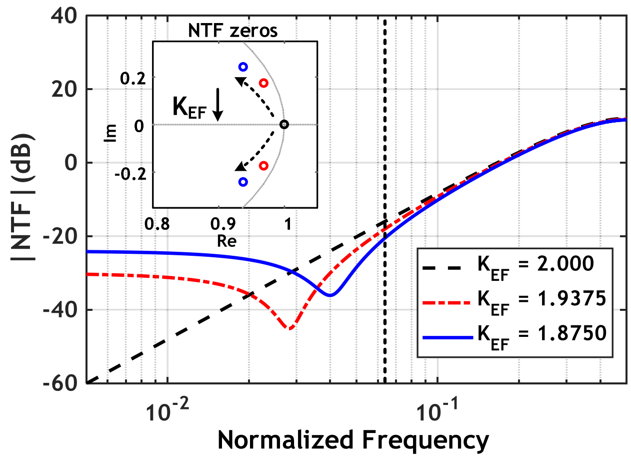

Fig. 5. Magnitude and zero locations of the NTF as a function of K_{EF}.

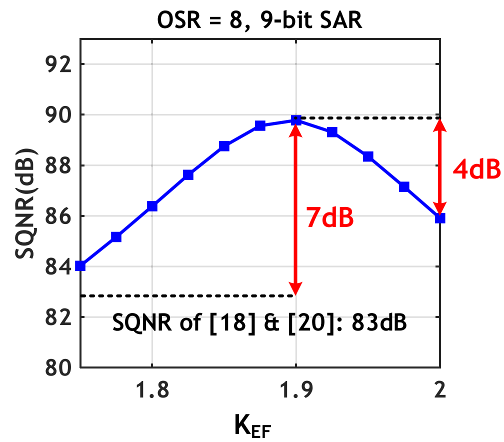

Fig. 6. Simulated SQNR as a function of K_{EF} with OSR = 8.

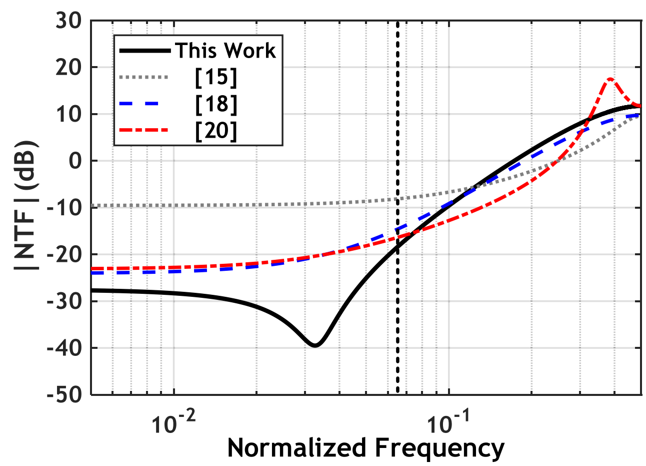

Fig. 7. Compare the NTF with [15], [18], and [20].

importantly, as A_{CS} is defined by capacitor ratios, the tuning nob of K_{EF} is only related to the D-Amp gain G. As can be seen in later discussion, G can be precisely controlled through the background loop. The NTF can, therefore, not only be accurately realized but also reconfigured flexibly to cater for different needs and fast redesign.

The change of the NTF magnitude and zero location versus K_{EF} is illustrated in Fig. 5. When $K_{\mathrm{EF}} = 2$, the two zeros are placed at $z = 1$, rendering the standard second-order high-pass NTF $(1-z^{-1})^2$. As K_{EF} reduces below 2, the NTF zeros become complex and move along a direction similar to the unit circle. Despite the complex zeros do not fall exactly on the unit circle, they are sufficiently close to it with the oversampling ratio (OSR) of 6 or beyond, thus able to create a strong notch that brings effective signal-to-quantization-noise-ratio (SQNR) improvement in the band of our interest. Fig. 6 shows the calculated SQNR as a function of K_{EF} assuming using a 9-bit SAR with OSR of 8. It shows a 4-dB SQNR boost compared to the zeros-at-dc condition ($K_{\mathrm{EF}} = 2$) when optimized. In Fig. 7, we can also horizontally compare the NTF with the state-of-the-art NS-SARs. Note that unlike prior works whose NTFs flatten out toward low frequency, this paper can always provide 2.5 b improvement per OSR doubling by adjusting the optimum zero location. This suggests that our design also well suits high-OSR high-resolution applications.

The optimum K_{EF} for a given OSR can be closely approximated as follows. When $K_{\mathrm{EF}} < 2$, the zeros of the NTF in (1) is given as:

$$Z|_{\mathrm{NTF}=0} = \frac{K_{\mathrm{EF}} \pm j\sqrt{2K_{\mathrm{EF}} - K_{\mathrm{EF}}^2}}{2}. \tag{2}$$

From (2), the magnitude of the zeros can be derived as $(K_{\mathrm{EF}}/2)^{1/2}$. This quantitatively explains the zeros' deviation from the unit circle. Yet, in the band of interest, the effect of the deviation is negligible, allowing us to safely treat the zeros as if they are on the unit circle, i.e., with the form $Z|_{\mathrm{NTF}=0} = e^{j\omega_z}$. Following the analysis in [23], for a second-order system, the optimum frequency of the zeros that minimized the in-band noise for a given OSR is given as

$$\omega_{z\_\mathrm{opt}} = \pm\frac{\pi}{\sqrt{3}\cdot \mathrm{OSR}}. \tag{3}$$

Associating (2) and (3), we have the following equation:

$$\arctan\left(\frac{\sqrt{2K_{\mathrm{EF\_opt}} - K_{\mathrm{EF\_opt}}^2}}{K_{\mathrm{EF\_opt}}}\right) = \frac{\pi}{\sqrt{3}\cdot \mathrm{OSR}}. \tag{4}$$

Since the input of the inverse tangent in (4) is close to 0, we can approximate the left-hand side with Taylor Series to the first order, which yields

$$K_{\mathrm{EF\_opt}} \approx \frac{2}{1+\frac{\pi^2}{3\mathrm{OSR}^2}}. \tag{5}$$

Fig. 8 shows the calculated $K_{\mathrm{EF\_opt}}$ as a function of OSR, using (5) and the strict approach by searching the minimum on the finite integral of (1). This comparison demonstrates the high accuracy of our approximation approach.

C. D-Amp Gain Consideration

Due to charge conservation, the charge sharing summation will bring attenuation to the input signal inside the capacitor

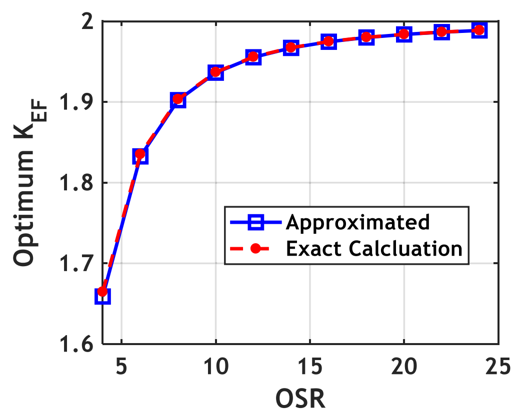

Fig. 8. Calculated $K_{\text{EF\_opt}}$ versus OSR based on (5) (blue) and exact calculation (red).

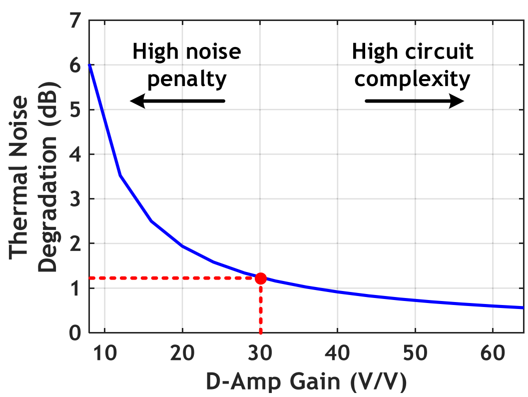

Fig. 9. Calculated thermal noise degradation as a function of D-Amp gain.

array by a factor of $1 - 2A_{\text{CS}}$. Note that this effect neither reduces the full-scale input range of the ADC nor degrades the sampling noise, because the charge sharing occurs after sampling and also attenuates the reference. However, it reduces the effective signal swing seen by the comparator input, making internal noise such as those from the comparator, D-Amp, and FIR filter more prominent, thus must be carefully circumvented by making A_{CS} sufficiently small. For a given K_{EF}, smaller A_{CS} necessitates higher G. Hence, there exists a tradeoff between the noise degradation and the D-Amp's complexity. To visualize it, Fig. 9 shows the calculated noise degradation (essentially noise figure) versus G using the sampling noise (kT/C_{DAC}) as a baseline. This graph only considers the noise from the FIR filter but not the comparator noise, because the latter is shaped and thus insignificant. A detailed noise analysis of this ADC is given in the Appendix. This paper chooses to use a D-Amp around 30 to balance the noise penalty and design difficulty. To realize this high gain efficiently, this paper adopts the technique to reuse the comparator as a D-Amp [25], which will be further described in the next section.

IV. Circuit Implementation

A. SAR Core and FIR

The top-level schematic and the timing diagram of the proposed EF NS-SAR ADC are shown in Fig. 10. For simplicity, the single-ended equivalent is shown. The SAR core adopts a 9-bit CDAC in the bottom-plate sampling fashion to attain high input linearity. It also employs the bidirectional single-sided switching technique to minimize the reference power [26]. The total single-ended capacitance of the DAC is 2 pF. This value is set to make the sampling noise contribution equal to the quantization noise. Under this capacitor size, nonetheless, the matching performance is insufficient. In this paper, the capacitor mismatches are mitigated through a one-time post-processing calibration [27], [28]. Timingwise, a cycle starts with processing previous conversion residues in the FIR and D-Amp (ϕ_{RST}, ϕ_{D2} and ϕ_{AMP}) to prepare the next feedback charges. After this commences the standard sampling and binary search, which takes up about the remaining 75% of a cycle. The EF summation (ϕ_{EF}) occurs during the last 5 b conversion for faster calibration. More explanation follows in Section C. Our comparator reusing technique obviates the need for an explicit D-Amp, making the passive SC FIR filter the only hardware required for the EF path.

The passive FIR consists of C_{res1}, C_{res2}, and C_{delay}, where C_{res1} serves as the one-cycle delay, C_{res2} and C_{delay} serve as the two-cycle delay with 0.5 attenuation. Its operation is detailedly depicted in Fig. 11. Note that the -1 gain is not an actual amplifier. It is implemented through cross-coupling in a fully differential setting. C_{res1}, C_{res2}, and C_{delay} have equal nominal capacitance of 143 fF, realizing $A_{\text{CS}} = 1/16$. In practice, the value of A_{CS} and the FIR coefficients will be affected by capacitor mismatch. The mismatch between C_{res1}, C_{res2}, and C_{delay} alters the 1:0.5 ratio between the two FIR taps, which will cause a shift in the NTF zeros and degrade the SQNR. To visualize the sensitivity, Fig. 12 shows the calculated SQNR as a function of the percentage ratio mismatch. It indicates that the degradation is less than 1 dB provided the mismatch stays within 2%. Since C_{res1}, C_{res2}, and C_{delay} are large capacitors and constructed closely with identical unit cells, their actual mismatch is expected to be less than 0.2% based on foundry data. Therefore, the SQNR degradation due to the mismatch between the FIR capacitors is negligible. The ratio mismatch between the SAR DAC and the FIR capacitors results in A_{CS} deviating from 1/16. However, this will be automatically compensated by the background calibration, which adjusts the D-Amp gain G accordingly to keep K_{EF} constant.

B. Comparator-Reused D-Amp

During the residue extraction, C_{res1} and C_{res2} are connected directly to the output of the comparator, which in this phase functions as a D-Amp [25], as shown in Fig. 13. The comparator-reused D-Amp exploits the combined integrating and the positive-feedback behavior of a strong-arm latch to achieve high gain efficiently in a single stage. Upon triggered, the nMOS transistors are first turned on and begins discharging the loading capacitor. At the same time, the input voltage difference will be linearly integrated on the load acting as preamplification. When the output common-mode drops low enough to turn on the pMOS pair, the regeneration phase takes over and further increases the gain exponentially to time.

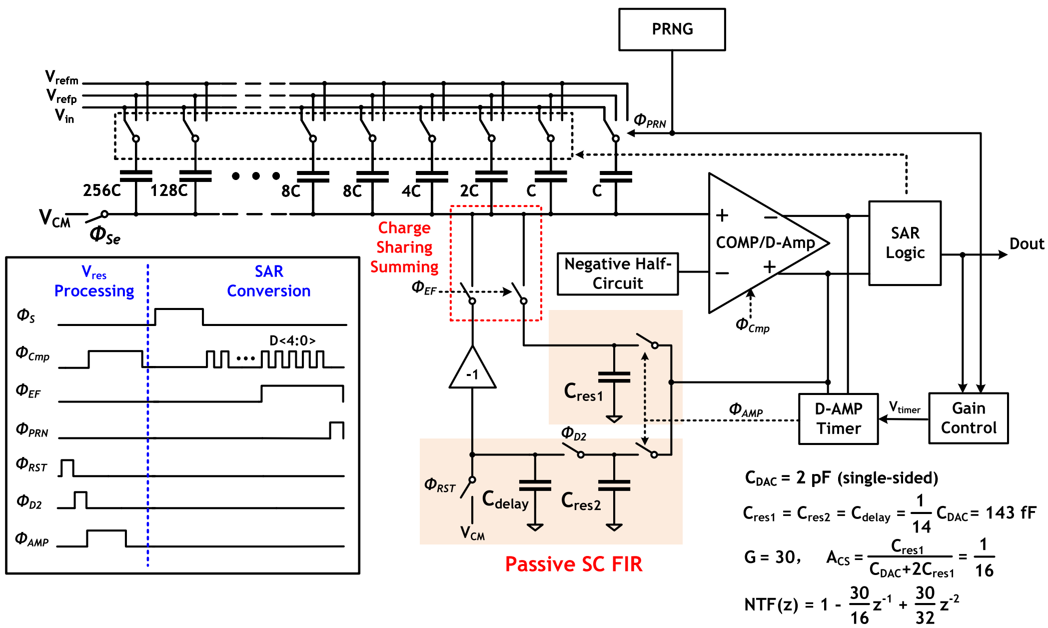

Fig. 10. Top-level schematic (single ended) of the proposed EF NS-SAR ADC.

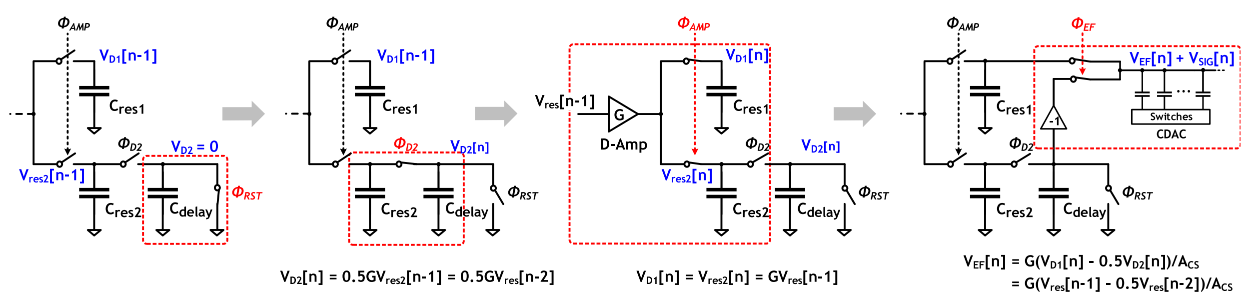

Fig. 11. Illustration of the FIR operation.

A timer signal will be asserted to disconnect the load from the latch before the latch fully regenerates. In this region, the output voltage of the comparator is proportional to its input, thus functions as an amplifier. Since the D-Amp only processes the residue of a 9-bit SAR, the gain is sufficiently linear. The noise of this D-Amp is dominated by the integration phase. By elongating the integration time, the input referred noise can be suppressed inverse proportionally [6], [30]. The loading from C_{res1} and C_{res2} naturally provides this bandwidth dampening when the comparator enters the amplification mode. Thus, the amplification can attain low noise without needing an over-designed comparator. As shown in Fig. 14, our D-Amp provides a gain of 30 within 2.5 ns with a low input referred noise of 85 μV_{rms}. Another advantage of the comparator-reused D-Amp is that the offset cancellation can be obviated, because the offset in the two modes is intrinsically matched.

Due to the exponential behavior of the comparator-reused D-Amp, the accurate timing of the load switch is critical to guarantee the D-amp's performance. Specifically, it must provide low timing jitter and the ability to track the gain curve over PVT variation. Timing jitter randomly shifts the shut-off point around its desired location, as illustrated in Fig. 14,

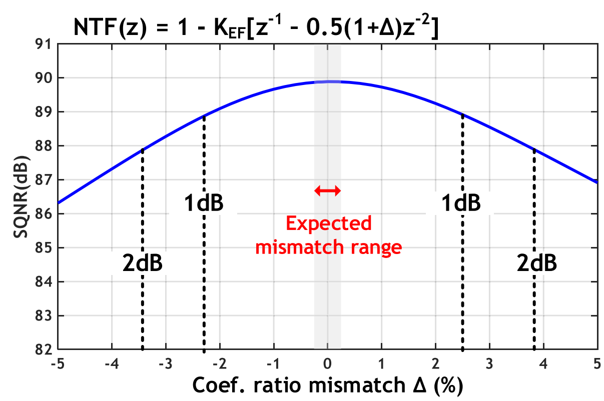

Fig. 12. Calculated SQNR versus percentage FIR coefficient ratio mismatch.

equivalently introducing a noise resulted from over or under amplification. To the first order, an instantaneous jitter-induced noise value can be approximated by the product of the output voltage derivative and the timing displacement. The output jitter-induced noise variance can therefore be expressed as

$$\sigma^2_{v_o\_\text{jitter}} = E\left\langle \left(\frac{dv_o(t_{\text{ideal}})}{dt}\right)^2 \right\rangle \sigma_j^2 \tag{6}$$

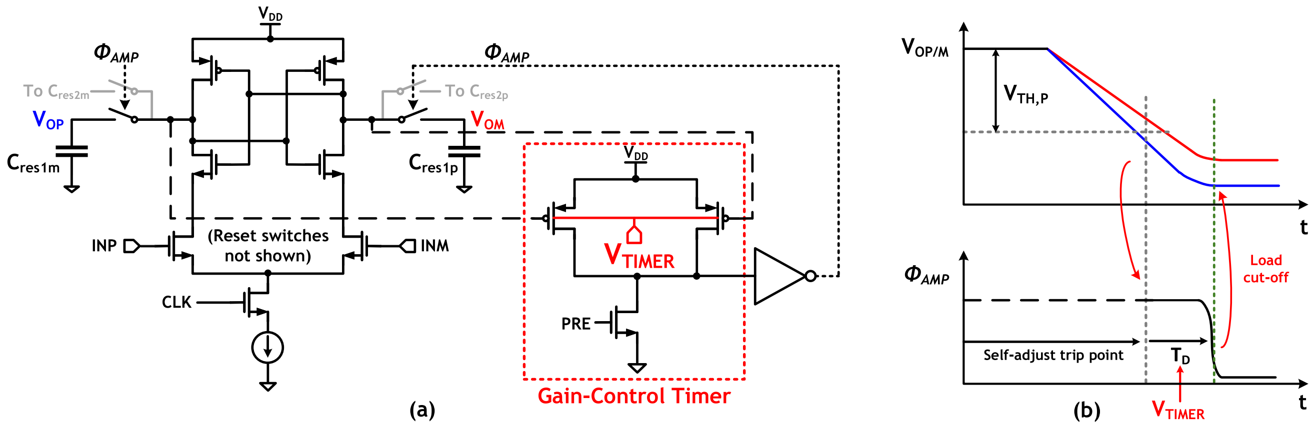

Fig. 13. (a) Schematic of the comparator-reused D-Amp and the self-trigger timer. (b) Operation principals of the timer.

where $\upsilon_o(t)$ is the D-Amp output voltage waveform, t_{ideal} is the nominal time for desired gain G_{ideal}, and σ_j is the root-mean-square (rms) jitter. During regeneration, the voltage curve can be approximated as

$$\upsilon_o(t) \approx \upsilon_{\text{res}} e^{t/\tau_r} \tag{7}$$

where υ_{res} is the D-Amp input (i.e., the residue voltage) and τ_r is the regeneration time constant. Equation (6) thus becomes

$$\sigma^2_{\upsilon_o\text{\_jitter}} = \frac{\sigma_j^2}{\tau_r^2} G^2_{\text{ideal}} E\langle \upsilon^2_{\text{res}} \rangle. \tag{8}$$

Equation (8) provides a guideline for the jitter upper bound. As τ_r, G_{ideal} are known design parameters and υ_{res} contains known quantization error and comparator noise, the rms jitter requirement can be readily derived for a given $\sigma^2_{\upsilon_o\text{\_jitter}}$.

To provide a PVT-robust D-Amp timing generation, this paper adopts a coarse–fine automatic gain-control scheme. The coarse tracking is implemented by a dynamic OR structure, as shown in Fig. 13, where the pMOS pair serves as a replica of the D-Amp's pMOS pair [29]. This self-triggered timer uses the D-Amp outputs' common-mode ramp in the integration phase as a reference, and produces a flip when the D-Amp enters regeneration. In other words, the timer can automatically adjust its trip point to follow the integration time variation, hence coarsely introducing a coherence between the timing and the gain curves. Another key advantage of this timer is that its short-delay path offers a very low jitter clock edge. The simulation shows that the rms jitter of the timer is 2 ps. Referring to (8), the resulted jitter-induced noise contributes to only 2% of the total D-Amp noise.

While the timer trip point tracks the integration phase, to ensure the eventual gain accuracy, a fine-tuning is also needed to match the timer delay with the regeneration phase variation. In this design, we use the back-gate bias of the timer's pMOS pair V_{TIMER} as the fine stage actuator. V_{TIMER} subtly modulates the strength of the pMOS pair to vary the delay of the timing path, thus possesses deterministic control over the D-Amp gain. The generation of V_{TIMER} is automated by a background calibration loop.

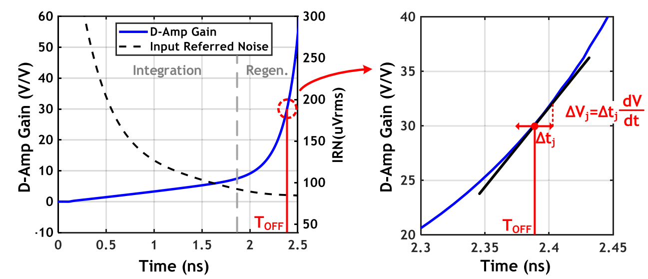

Fig. 14. Time-domain behavior of the comparator-reused D-Amp and its jitter effect.

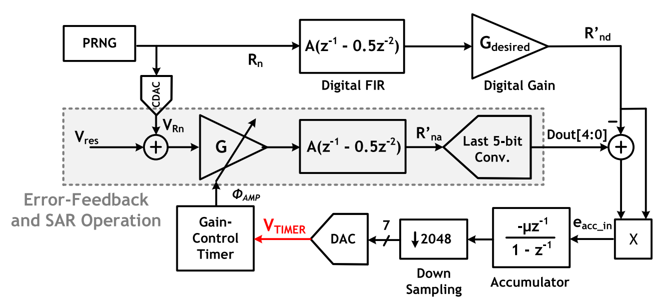

Fig. 15. Logic flow diagram of the D-Amp gain control background calibration.

C. Background Calibration Loop

The logic flow diagram of the proposed gain-control background calibration scheme is shown in Fig. 15. To help understand the calibration, we can view the logic flow as synchronously driven by only one clock (e.g., ϕ_{PRN}) and treat delay (z^{-1}) as D-flip-flops while the rest simply as combinational gates. The calibration is based on the dither-based least-mean-square (LMS) algorithm. At the end of each SAR conversion, it injects a binary pseudo-random number (PRN) generated from an on-chip linear-feedback shift register (LFSR) into the conversion residue through the SAR CDAC. The PRN will be processed by the EF path and appear at the output codes of subsequent conversions. Meanwhile, the same PRN will also go through a digitally represented "desired" EF path. This digitally filtered PRN

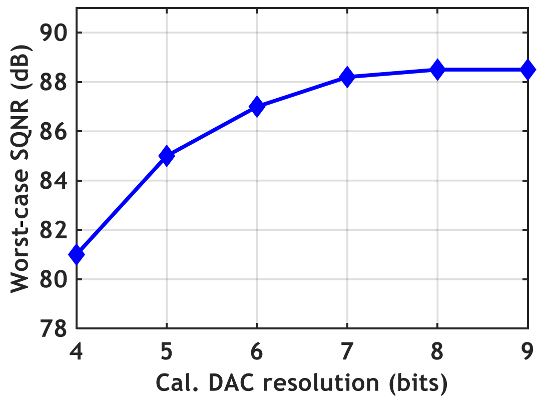

Fig. 16. Simulated worst case SQNR versus calibration DAC resolution.

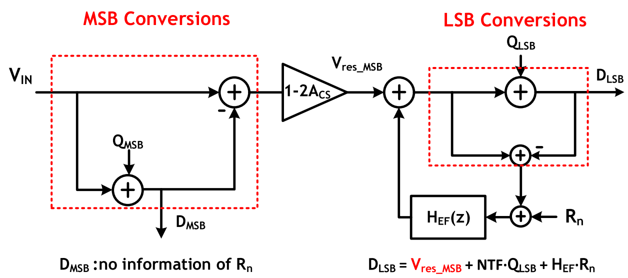

Fig. 17. EF during LSB conversion leverages the idea of subranging.

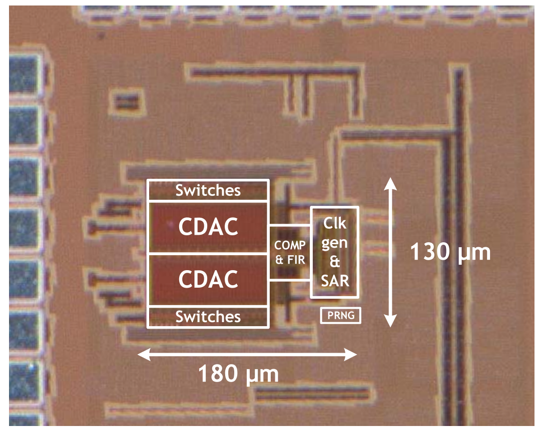

Fig. 18. Die photograph.

will then multiply with the difference between itself and the ADC output. The multiplication results are accumulated to drive V_{TIMER} to actuate gain change. The accumulator will only settle when the dc-level of its input reaches zero. Mathematically, this corresponds to the following equation:

$$\overline{e_{acc\_in}} = \overline{(R'_{na} + D_{sig} - R'_{nd})R'_{nd}} = 0 \quad (9)$$

where R'_{na} and R'_{nd} are the analogly/digitally filtered PRN, respectively, and D_{sig} is the ADC output that contains only the signal and the system noise. By design, D_{sig} is uncorrelated with the PRN, i.e., $\overline{D_{sig}R'_{nd}} = 0$. Therefore, (9) is satisfied when $R'_{na} = R'_{nd}$, which can only be attained when $G = G_{desired}$, or more precisely, $K_{EF} = K_{EF\_desired}$. This describes how the background loop regulates the D-Amp gain. Note that the above-mentioned analysis assumes the analog and digital paths are well matched. In practice, mismatches in the analog components (dither capacitor, FIR capacitors, etc.) affect the accuracy of the calibrated gain. Nevertheless, this is deemed non-critical as we can guarantee good matching easily (within 1% deviation) in the proposed EF structure, thanks to its simple structure and ratio defined coefficients.

The proposed background calibration is mostly digital and can be implemented efficiently without using any full-blown multiplier. The digital gain can be implemented using a shift-and-add manner. For example, the gain of 30 can be realized by 32× subtracting 2×. The error multiplier can also be implemented simply through multiplexing, shifting, and adding. This is because, R'_{nd} only has four possible values $[(+1.5, +0.5, -0.5, -1.5) \times K_{EF\_desired}]$ due to the PRN being binary. The DAC used for converting the accumulator output to V_{TIMER} also has very relaxed resolution and speed requirements. As shown in Fig. 16, a 7 b DAC in the range of 0.6–1.1 V is sufficient to suppress the truncation impurity. As the accumulation limits the bandwidth of the background loop, the DAC can be updated very infrequently, such as 1/2048 of the ADC rate in our case, without hindering the loop performance. The calibration logic was synthesized and simulated with a 2-MΩ resistor string DAC. The digital portion consists of only 25 full adders, 12 half adders, 40 D-flip-flops, and 30 simple combinational gates. The estimated area overhead of the calibration module including the DAC is less than 20% of the ADC core area. The simulated power of the consumption of the module is 4 μW, which is only 5% of the ADC core power.

To speed up the calibration convergence, this paper is designed to apply the EF summation only during the last 5 bits (4 LSB + 1 b redundancy) of the SAR conversion. This arrangement leverages the fact that the SAR ADC is also a subranging ADC, as illustrated in Fig. 17. By doing so, we can restrict the PRN to correlate with only the relatively small noiselike MSB residue rather than the entire full swing signal, hence reducing the interference from the signal components during the gain error accumulation. On the top of accelerating, this technique also attenuates any possible correlation in the ADC input with the PRN, making the calibration more robust against signal type. With 8-LSB redundancy, the back-end conversion is protected from being overloaded by the EF injection and able to tolerate large initial K_{EF} up to 2.5× the nominal. While allocating fewer bits for the back-end stage is an option to further accelerate convergence, it should be aware that this may necessitate extra cycles to cover the redundancy range when the back-end range is less than the redundancy.

During startup, the calibration DAC will be initialized to the mid-scale code. While the time required to settle to the working code depends on PVT variations, a worst case estimation can be obtained by assuming the DAC settles to its boundary. Simulation reveals the worst case startup requires about 1.2-M cycles.

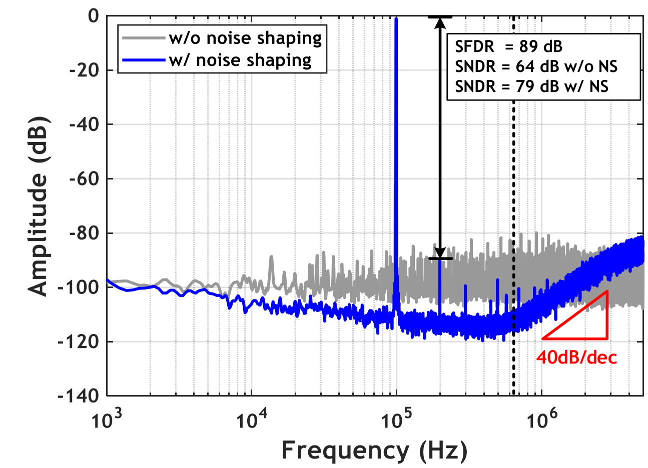

Fig. 19. Measured output spectrum with noise-shaping enabled/disabled.

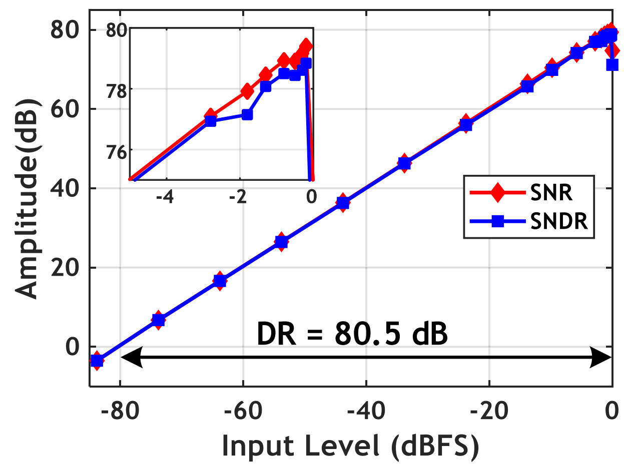

Fig. 20. Simulated SNDR/SNR versus input signal magnitude.

V. Measurement Results

Fig. 18 shows the silicon die photograph of the prototype NS-SAR ADC fabricated in 40-nm LP-CMOS process. The design occupies an active area of 0.024 mm^2, among which less than 10% is contributed by the EF path (i.e., the comparator-reused D-Amp and the passive SC FIR). This testifies that the hardware overhead needed to realize the optimized noise shaping is low.

Under 1.1-V supply, the prototype NS-SAR consumes 84 μW when sampling at 10 MHz, where 21, 29, and 34 μW are contributed by analog, reference, and digital, respectively. Out of the 21-μW analog power, the strong-arm latch consumes 14 μW for comparison and 6 μW as D-Amp, with the rest used by driving the CDAC switches. Fig. 19 shows the measured output spectra with noise-shaping disabled/enabled. The peak signal to noise and distortion ratio (SNDR) improves from 64 to 79 dB over 625-kHz bandwidth when noise shaping is on, with a clear second-order high-pass shaping. Thanks to the optimized NTF, this paper essentially achieves 13 effective number of bits (ENOB) with a relatively low OSR of 8. As mentioned previously, the CDAC mismatches are mitigated through a one-time calibration. This calibration is performed by feeding in a low-frequency sinusoidal test signal to excite all the bits, then applying least squares regression to figure out the optimal capacitor weights that yield the lowest harmonic distortion [27], [28]. The set of weights will then be frozen and applied to all subsequent tests. While the sinusoidal input and the parameter fitting are adopted for simplicity, the proposed ADC is compatible with existing mismatch calibration techniques that have no input restrictions [31]. The explanation can be found in the Appendix. The calibrated spurious-free dynamic range (SFDR) reaches 89 dB, with the second-order harmonic being the limiting tone. We thereby infer that these remaining spurious tones are originated from the signal path and the sampling network. The measured SNR/SNDR versus input amplitude is presented in Fig. 20, showing a DR of 80.5 dB. The ADC's performance is consistent for different in-band input frequencies.

The D-Amp gain control background loop is validated off-chip through the setup depicted in Fig. 21 during the measurement. The centerpiece of the calibration is implemented in MATLAB. It includes an m-script that exactly models and runs the calibration operations in Fig. 15, as well as instrument utilities for passing the data. It closes the loop by continuously processing the NS-SAR output captured by the logic analyzer and adjusting the value of V_{TIMER} through the on-board calibration DAC,[1] thus realizing the desired on-the-fly gain regulation. Note that the loop actually runs at a reduced real-time fashion, in a sense that the loop will be disabled intermittently to allow the instruments to communicate or reset. However, this does not affect accuracy since no information is lost. During measurement, the gain of the D-Amp/EF-path can be extracted by directly multiplying the ADC output with the PRN sequence and perform an averaging [32]. To verify the robustness over temperature, a three-step temperature sweep is performed, as shown in Fig. 22. The settled SNDR exhibits less than 1-dB variation, demonstrating the effectiveness of the proposed calibration scheme. Owing to the reduced real-time constrain mentioned above, the off-chip loop time constant becomes longer than the temperature ramping speed. Hence, SNDR droops are observed at the beginning of each transition. However, this is not likely to occur for actual applications with on-chip background loop. The measured tuning range for V_{TIMER} is 140 mV for covering 80 °C. Fig. 23 reveals the time constant, which is obtained by

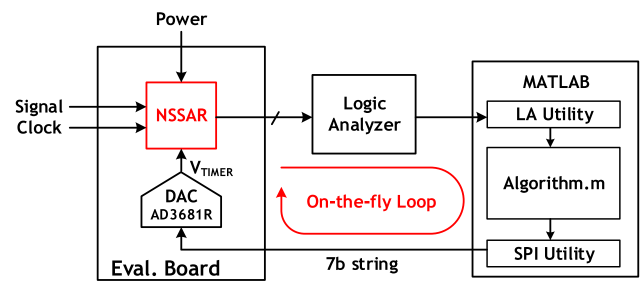

Fig. 21. Off-chip background calibration implementation.

[1] Although AD3681R is a 2.5-V 12 b DAC, we deliberately truncate our DAC control to utilize only 7 bits in the range of 0.6–1.1 V to match our intended on-chip implementation.

TABLE I
PERFORMANCE SUMMARY AND COMPARISON WITH STATE-OF-THE-ART NS-SAR ADC

Specifications	[12]	[13]	[14]	[15]	[18]	[20]	[21]	**This work**
Architecture	CIFF	CIFF	CIFF	Pseudo-EF	CIFF	CIFF	CIFF	**True EF**
Opamp-free	✗	✗	✗	✓	✓	✓	✓	✓
Optimized NTF Zeros	✗	✗	✗	✗	✗	✗	✗	✓
PVT Robust	✓	✓	✓	✓	✓	✗	✗	✓
Technology [nm]	65	55	28	65	40	28	65	**40**
Active Area [mm^2]	0.03	0.072	0.116	0.012	0.04	0.0049	0.08	**0.024**†
Supply [v]	1.2	1.2	1.8/1.1	0.8	1.1	1	1	**1.1**
Fs [MS/S]	90	1	1	50	8.4	132	25	**10**
Bandwidth [kHz]	11000	1	20	6250	262	5000	625	**625**
OSR	4	500	25	4	16	13.2	20	**8**
SNDR [dB]	62	101	94	58	80	80	80	**79**
Power [uW]	806	15.7	493	121	143	460	630	**84**†
FoMs* [dB]	163	179	170	165	173	180	170.4	**178**
FoMw** [fJ/step]	36	85	305	15	33	5.8	61	**9**

**FoMs = SNDR + 10*log10(BW/Power)*
***FoMw = Power/(2^{ENOB}*2*BW)*

† *Area and power numbers include on-chip PRNG, but exclude the estimation for the rest of calibration*

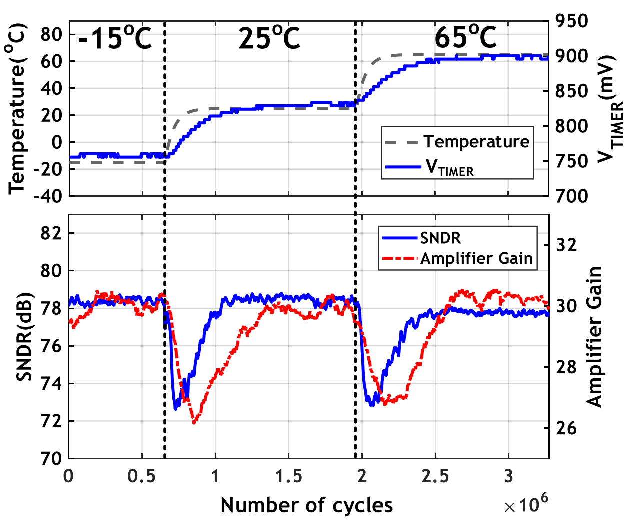

Fig. 22. Measured timer control voltage, SNDR, and D-Amp gain versus temperature.

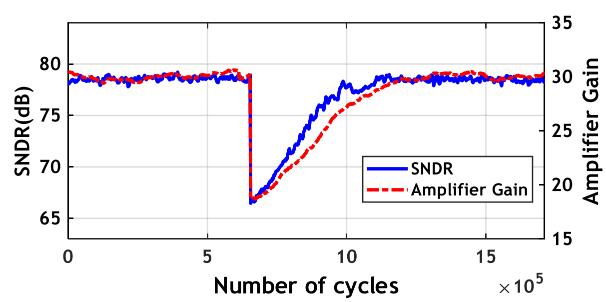

Fig. 23. Measured voltage supply step response of the background loop.

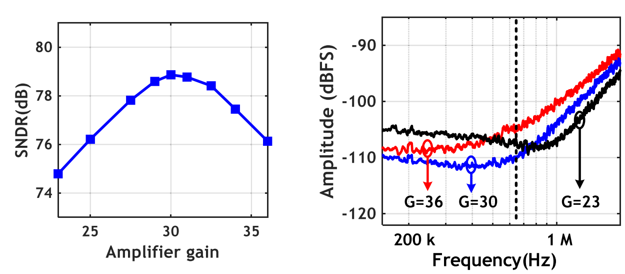

Fig. 24. Measured SNDR and NTF profile as a function of D-Amp gain.

stimulating the ADC with a 5% voltage supply step, to be about 300-k cycles. This translates to 30 ms for a true real-time loop, which is sufficient to track most real-time changes. The measured SNDR and PSD under various D-Amp gain settings are shown in Fig. 24. The peak SNDR is achieved with at D-Amp gain of 30; it corresponds to a K_{EF} of 1.875. Its corresponding measured PSD also suggests a lowest in-band noise floor. The measured optimum gain setting is in good agreement with the analysis.

A performance summary and a comparison over existing NS-SAR ADCs are given in Table I. This design achieves a peak Schreier and Walden FoM of 178 dB and 9 fJ/step, respectively, demonstrating in-line performance with state of the arts. To the authors' best knowledge, our work stands out for being the first NS-SAR to realize optimized NTF while maintaining a low-power OTA-less implementation. It also sustains PVT robustness, though employing D-Amp, by devising a background calibration scheme.

VI. CONCLUSION

This paper presented a second-order NS-SAR ADC that leverages the EF structure. With a reduced variable EF scheme, it can realize complex NTF zeros for effective noise-shaping enhancement with a minimum modification to a SAR ADC.

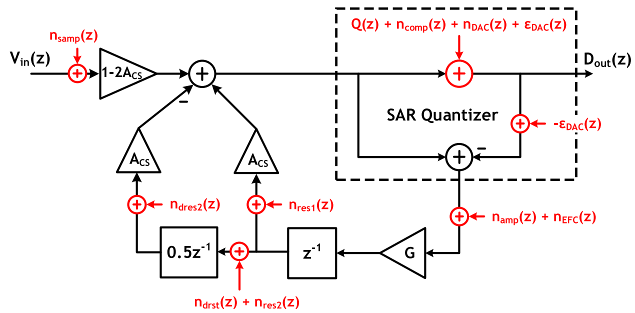

Fig. 25. Noise effects in the proposed EF NS-SAR.

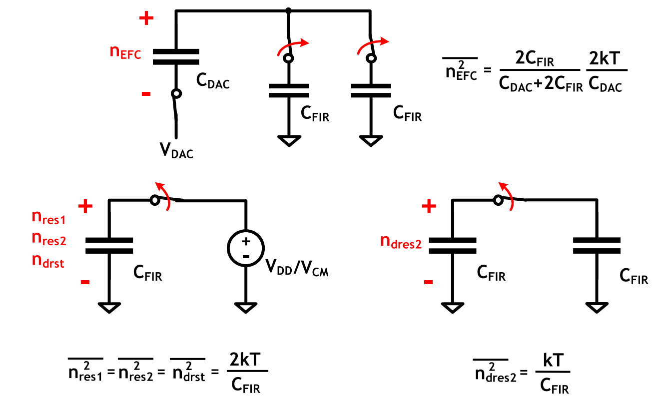

Fig. 26. Derivation of the noise in the SC FIR operation.

A low-cost background calibration scheme is also proposed to maintain the ADC's PVT robustness. This paper has demonstrated an effective solution to realize NTF optimization for NS-SAR ADCs with simple low-power scaling-friendly circuits. It also proves the strong potential of the EF structure for implementing high-efficiency NS-SAR ADCs. The proposed techniques can be easily reconfigured, enabling a flexible and low-power solution for IoT applications.

Appendix
Noise Analysis of the Proposed ADC

The non-ideality and noise effects of the proposed EF NS-SAR are investigated in Fig. 25. The total output noise can be expressed as

$$\begin{aligned} n_{o\_\text{total}} = {} & \text{NTF}(z)[Q(z) + n_{\text{comp}}(z) + n_{\text{DAC}}(z)] \\ & + (1 - 2A_{\text{CS}})n_{\text{samp}}(z) + \varepsilon_{\text{DAC}}(z) \\ & + H_{n1}(z)[n_{\text{amp}}(z) + n_{\text{EFC}}(z)] \\ & + H_{n2}(z)[n_{\text{res1}}(z) + n_{\text{dres2}}(z)] \\ & + H_{n3}(z)[n_{\text{drst}}(z) + n_{\text{res2}}(z)] \end{aligned} \tag{10}$$

where the transfer function $H_{n1}(z)$, $H_{n2}(z)$, and $H_{n3}(z)$ are

$$H_{n1}(z) = K_{\text{EF}}(z^{-1} - 0.5z^{-2}) \tag{11}$$

$$H_{n2}(z) = A_{\text{CS}} \tag{12}$$

$$H_{n3}(z) = 0.5A_{\text{CS}}z^{-1}. \tag{13}$$

Q, n_{comp}, and n_{DAC} represent the quantization error, comparator input referred noise, and DAC noise, respectively. These noise will be second-order shaped by the NTF of the system, therefore imposing insignificant effect to the total noise contribution. n_{samp} is the differential input sampling noise given by $(2kT/C_{\text{DAC}})$. ε_{DAC} is the DAC nonlinearity. The non-linearity appears at the output directly without modification. This suggests that available nonlinearity calibrations for standard SAR ADC, dynamic element matching (DEM), or mismatch shaping techniques can be applied directly in this paper. n_{amp} is the input referred noise of the D-Amp. As a detail derivation has been reported in [33], we will skip its discussion here. The D-Amp input referred noise can be expressed as

$$\overline{n_{\text{amp}}^2} = \left(1 + \frac{1}{2G_{\text{int}}}\right)\frac{2kT\gamma}{C_{\text{FIR}}G_{\text{int}}} \tag{14}$$

where G_{int} is the integration phase gain, γ is the noise factor, and C_{FIR} is the capacitance of an FIR tap.

n_{EFC}, n_{res1}, n_{res2}, n_{drst}, and n_{dres2} are related to the SC operations in the EF path. n_{EFC} captures the noise sampled on the CDAC when disconnecting the FIR capacitors before the residue amplification. n_{res1} and n_{res2} are the sampled noise on C_{res1} and C_{res2}, respectively, during D-Amp pre-charge. n_{drst} signifies the noise sampled on C_{delay} during its reset, and n_{dres2} is the additive noise during the charge transfer from C_{res2} to C_{delay}. The derivation of their noise power is shown in Fig. 26. This analysis serve as a guideline for noise budgeting as well as choosing the D-Amp gain during the design of this paper. For the readers' interests, the noise budgeting of this paper is shown in Table II.

TABLE II
Noise Budgeting of the Proposed NS-SAR

	RMS Voltage	Percentage
Sampling	20 μV_{rms}	14%
Quantization	21 μV_{rms}	14%
SC FIR	11 μV_{rms}	4%
D-Amp	45 μV_{rms}	68%

Acknowledgment

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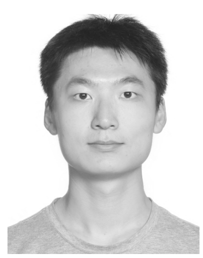

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